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Posted by u/Ishannaik 1 year ago

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A Clear roadmap to complete learning AI/ML by the end of 2022 from ZERO

Request

I've always been a tech enthusiast since I was a Kid I'm 18 now and I always wanted to learn how it works and make it myself, I've got myself into a good college but had to sacrifice my branch of bachelor in computers and choose electronics (because my score wasn't enough), I wish to learn but I do not have any clarity on where to start and where to go what I'm looking for is to pursue a degree in CS masters but I'll have to learn everything by myself so if any of you have a clear roadmap please let me know

37 Comments Share ••• 92% Upvoted

What are your thoughts?

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MarcelDeSutter · 1 yr. ago 2

Let me cite one of my own comments, since it fits perfectly I think:

Hi :) So I suppose I'm infamous on this subreddit for providing very lengthy roadmaps to learning ML: https://www.reddit.com/r/learnmachinelearning/comments/cxrpjz/a_clear_roadmap_for_mldl/eyn8cna

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My plan is to present these information in a more professional manner, i.e. on YouTube. But I just saw your post and I thought 'why not post my notes for this video I'm planning on this matter?' So here you go, consider this a sneak peak ;)

Level 1 – Informed Decision Maker:

- Basic understanding of what ML is and what it is not
- Know how ML can/will affect their lives in the short to mid term
- Know how ML's potential can be utilized to serve themselves (or their teams)

resources:

```
coursera – ai for everyone
andrew ng – machine learning yearning
coursera – machine learning (first three weeks
100 page ML book
```

From now on, three areas of focus will be given for each level: Mathematics, Concrete ML knowledge, and Programming

Level 2 – Competent Developer

- Have basic intuition about the math relevant for ML
- Know the theory behind the most basic ML algorithms (linear/logistic regression, svm, decision trees/random forests, knn clustering, basic neural networks)
- Know the basics of the Python programming language, the data science stack (numpy, pandas, matplotlib/seaborn, sklearn, sql queries) and how to implement basic ml pipelines

Mathematics:

```
Linear Algebra:
        - Gilbert Strang – MIT online lecture
        - 3blue1brown – Essence of linear algel

(Multivariate) Calculus:
        - 3blue1brown – Essence of calculus
        - Khan Academy – AP/College Calculus Al
        - Khan Academy – Multivariable calculu

Statistics and Probability:
        - Khan Academy – Statistics and Probab
```

Concrete ML Knowledge:

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```
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```

Programming:

```
Corey Schafer – Python Programming Beginner Tu
Corey Schafer – Python OOP Tutorials – Working
Corey Schafer – Jupyter Notebook Tutorial
Corey Schafer – Pandas Tutorials
Corey Schafer – Matplotlib
Kaggle – Microcourses
Keith Galli - Complete Python NumPy Tutorial
Streamlit
data science handbook (a bit verbose for self
WQU - Applied Data Science Module:
        - Applied Data Science I: Scientific C
        - Applied Data Science II: Machine Lea
```

Level 3 – Expert Developer

- Have enough mathematical proficiency to be able to read academic papers or graduate level textbooks about ML
- Have extensive knowledge and understanding of a wide range of ML algorithms to be able to apply the correct the algorithm for the problem at hand. Be able to discuss the pros and cons of different algorithms and consult decision makers
- Know how to address the challenges of dealing with stochastic code and be able to create complex ml pipelines that can be integrated into larger software infrastructures

Mathematics:

```
Statistics and Probability:
        - MITx – Probability-The Science of Un
        - MITx – Fundamentals of Statistics

Wide range of topics:
        - Ulrike von Luxburg – Mathematics of
```

Concrete ML Knowledge:

```
- Kilian Weinberger: Machine Learning for inte
- Andreas Geiger: Deep Learning
- Ulrike von Luxburg: Statistical Machine Lear
```

Programming:

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```
- Jose Portilla (Udemy): NLP - Natural Languag
- fast.ai
- d2l
- Soledad Galli:
        - deployment of machine learning model
        - feature engineering for machine lear
        - feature selection for machine learni
        - testing and monitoring machine learn
        - machine learning with imbalanced dat
- Refactoring Guru Design Patterns
- udacity courses
```

Level 4 – PhD level

- Deepen understanding of advanced mathematics and selected branches of ML to be able to read exotic/very theoretical papers, perhaps even contribute by creating theoretical insights on your own
- Be able to contribute to open-source projects and create innovative software products yourself

The resources for this level are more free-form, depending on your specialization:

```
textbooks

papers with code
fast.ai 2
fast.ai code first introduction to nlp
fast.ai numerical linear algebra

AMMI - Geometric Deep Learning Course
steve burton - machine learning and dynamical
tübingen – probabilistic machine learning
tübingen – computer vision
penn university – graph neural networks
stanford – reinforcement learning
deepmind – introduction to reinforcement learn
stanford – natural language processing with de
openmined – private ai series

machine learning street talk
lex friedman (not terribly rigorous but inspir
```

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1shannak OP · 1 yr ago

Thanks, A lot I'll keep updating this comment to hold myself accountable and track what I've completed till now

(And of course my YouTube channel: https://www.youtube.com/channel/UCg5yxN5N4Yup9dP_uN69vEQ)

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Level 1 – Informed Decision Maker:

~~coursera – ai for everyone~~

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coursera - machine learning (first three weeks)

100 page ML book

Level 2 – Competent Developer

Mathematics:

Linear Algebra:

`- Gilbert Strang – MIT online lecture (find problems and solutions)`

`- 3blue1brown – Essence of linear algebra`

(Multivariate) Calculus:

~~- 3blue1brown – Essence of calculus~~

`- Khan Academy – AP/College Calculus AB`

`- Khan Academy – Multivariable calculus`

Statistics and Probability:

`- Khan Academy – Statistics and Probability`

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a1b3rt · 8 mo. ago

This is a great resource, bookmarked!

One question for you --

If someone is a Cloud Solutions Architect (Azure/GCP/AWS) and wishes to be an expert on AI/ML solutions -- I reckon they would not need the "Expert Developer" level proficiency on building hands-on ML/AI applications but need to have enough knowledge to architect and design for these use cases.

Do you have any thoughts on what would be a good set of skills and topics and what depth to pursue for this persona?

Thank you!

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Zeebo42X · 6 mo. ago

Commenting for follow up :)

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> Thanks a lot for this guide , I will update my comment when I finish a course to keep myself accountable
>
> **Level 1 – Informed Decision Maker**
>
> ~~coursera – ai for everyone~~
> andrew ng – machine learning yearning
> coursera – machine learning (first three weeks)
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Statistics and Probability:

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\- Khan Academy – Statistics and Probability
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> Concrete ML Knowledge:

coursera – machine learning

coursera – deep learning specialization (courses 1 to 4 on youtube)

Dmitry Kobak – introduction to machine learning

> Programming:

Corey Schafer – Python Programming Beginner Tutorials

Corey Schafer – Python OOP Tutorials – Working with Classes

Corey Schafer – Jupyter Notebook Tutorial

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data science handbook (a bit verbose for self study)

WQU - Applied Data Science Module:

```
\- Applied Data Science I: Scientific Comput

\- Applied Data Science II: Machine Learning
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Illusion_and_Dream · 1 yr. ago

Im a Master Degree student in AI and cognitive science that has almost completed his studies.
Back in 2020 I was a little bit "lost" in this huge and, for me, marvellous world but I have found some resources of Daniel Bourke. It's a youtuber that does almost exclusively programming videos. The one that caught me was his roadmap to become a self-thought ML engeneer.
https://www.mrdbourke.com/2020-machine-learning-roadmap/

This is a post on his personal website where he shows ALL the things that he had studied to become what he is now.
The BEST thing is that HUGE roadmap where he show ALL the books he had studied and main topics to cover starting from plain algebra, matrices, python courses for beginners till Machine Learning and Deep Learning notions.

It's a hard roadmap that probably will take more than a year to complete with full knowledge of everything (now I know most of the things that he showed here but I have done 2 years of University only on that with projects and papers) but it is a very good starting point and, if you want, you could even deep dive in concepts that he just mentions.

Let me know your thought about that.

85 Reply Share Report Save Follow

Ishannaik OP · 1 yr. ago

> https://www.mrdbourke.com/2020-machine-learning-roadmap/
>
> This is a post on his personal website where he shows ALL the things that he had studied to become what he is now.

Thanks a lot, I'll use this as my reference to learn about this stuff

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wickedandwindy · 1 yr. ago

Thanks, this looks great.

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Pirate_Assassin_Spy · 2 mo. ago

> https://www.mrdbourke.com/2020-machine-learning-roadmap/

Hey, can I ask where you're studying? This is the exact Masters I want to do! Thanks so much.

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amitness · 1 yr. ago

Not exactly a roadmap, but I documented my learning journey here: https://github.com/amitness/learning

It's been 3+ years + a full-time ML job and I still feel there is so much to learn.

I think Daniel Bourke's roadmap mentioned in the other comment is very relevant for someone starting their journey.

7 Reply Share Report Save Follow

d0r1h · 1 yr. ago

Don't know how much it helps but there are plenty of resources online which are very good, but it's hard to keep up with them, so keeping this thing in mind, I developed this repo where you can find all the best course on the internet for free.

https://github.com/d0r1h/ML-University

I'll continue update with the new course and important resource that can help someone in their journey.

So just take a course and head start learning :)

6 Reply Share Report Save Follow

aba1476 · 1 yr. ago

I have a different perspective. What really helped me in getting the end to end picture is realizing ML in real life and that includes Data Ingestion, Data engineering (believe it or not - these take the most of the time), then ML modeling (the comments earlier provide a very good list) and not to forget deploying the models and

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Do some intro courses on Python ML (classification/regression) models.

Note: In most cases folks use XGBoost/ Random forests - they are the general-purpose soup that fits well in most cases. Lately, the flavor of the day is deep learning models.

I would recommend you dive deep into AWS SageMaker or Azure ML or Google Cloud ML. pick 1 of the 3 and follow the guide cover to cover (the docs provided are excellent) and they will have a gazillion notebooks on Git that you can just use to understand. I used SageMaker and did the ML certification exam. This path will indeed help you in real life and that's what the market is seeking.

Hope this helps.

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[deleted] · 1 yr. ago

Learn statistics, and then from linear algebra learn matrix multiplication and the eigenvalue problem. Also reading a chapter on subspaces and linear transformations wouldn't be bad to.

Then decide if you want to learn Machine Learning (Really this is just using classical statistical models on datasets) or you can learn Deep Learning (I find this more interesting)

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Ishannaik OP · 1 yr. ago

I'm not quite sure about the difference between DeepLearning And Machine Learning is yet I'll check it out

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TheManWithNoDrive · 1 yr. ago

Hey Op, as a side note, I'm not sure which country you're from and how this may apply, but maybe you can see what you can do to transfer over to CS now that you're in a close field as it is (where I'm from, this would all be under the term "Engineering" as the college of engineering would handle this within the university"). It might be easier to transfer in than it was to originally join. That might help with you getting the classes and not overwhelming yourself with a lot of information

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Ishannaik OP · 1 yr. ago

Not really I'm from India and education is just another form of business here there will be no way to transfer except paying tons of money as "donations" and yes I'm in a college of engineering

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Ne_oL · 1 yr. ago

Check fast.ai courses, part 1 (8 lectures) and part 2 (i think also 8). They would set you with the basics, ethics, and general comprehension of the field, not to mention their awesome (though a bit too easy API).

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newjeison · 1 yr. ago

Focus on your undergrad and look for research opportunities. It's hard learning about the math behind ML/AI on your own.

1 Reply Share Report Save Follow

saebeas · 7 mo. ago

Amazing thank you, bookmarked.

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NeutralFan123 · 11 days ago

Hi if I want to have enough knowledge to make my own chatbot that uses AI and ML capabilities how proficient will my knowledge have to be based on this roadmap

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sidhanti · 1 yr. ago

Remind me

0 Reply Share Report Save Follow

wadaphunk · 1 mo. ago

Kind reminder

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